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root@ip-172-31-10-233:~# #Lab57:Simple_Log_Rotation
root@ip-172-31-10-233:~# #OBJECTIVES: Understand the basics of log rotation in a Linux
environment. Learn how to inspect and modify configuration files for log rotation. Gain hands-on
experience in manually initiating log rotation.
root@ip-172-31-10-233:~#
root@ip-172-31-10-233:~# #Task1:Check default configuration in /etc/logrotate.conf
root@ip-172-31-10-233:~# cat /etc/logrotate.conf
# see "man logrotate" for details
```

```
# global options do not affect preceding include directives
```

```
# rotate log files weekly
weekly
```

```
# use the adm group by default, since this is the owning group
# of /var/log/.
su root adm
```

```
# keep 4 weeks worth of backlogs
rotate 4
```

```
# create new (empty) log files after rotating old ones
create
```

```
# use date as a suffix of the rotated file
#dateext
```

```
# uncomment this if you want your log files compressed
#compress
```

```
# packages drop log rotation information into this directory
include /etc/logrotate.d
```

```
# system-specific logs may also be configured here.
```

```
root@ip-172-31-10-233:~# sudo logrotate -f /etc/logrotate.conf
```

```
root@ip-172-31-10-233:~# #weekly: Rotates the log files once every week (based on calendar
week).
```

```
root@ip-172-31-10-233:~# #rotate 4: Keeps only the last 4 rotated (old) versions of the log before
deleting the oldest one.
```

```
root@ip-172-31-10-233:~# #create: After rotation, creates a new empty log file with the original
name so the service can continue writing seamlessly.
```

```
root@ip-172-31-10-233:~#
```

```
root@ip-172-31-10-233:~# #Task 2: Inspect a File in /etc/logrotate.d/
```

```
root@ip-172-31-10-233:~# cd /etc/logrotate.d
```

```
root@ip-172-31-10-233:/etc/logrotate.d# ls
```

```
alternatives apt      btmp    cloud-init  dpkg  rsyslog    speech-dispatcher  ubuntu-pro-
client  unattended-upgrades
```

```

apport      bootlog chrony cups-daemon ppp sane-utils sssd-common      ufw
wtmp
root@ip-172-31-10-233:/etc/logrotate.d# cat bootlog
/var/log/boot.log
{
    missingok
    daily
    copytruncate
    rotate 7
    notifempty
}
root@ip-172-31-10-233:/etc/logrotate.d# #missingok: skip errors if log file is missing
root@ip-172-31-10-233:/etc/logrotate.d# #notifempty: do not rotate empty log files.
root@ip-172-31-10-233:/etc/logrotate.d#
root@ip-172-31-10-233:/etc/logrotate.d# #Task 3: Manually Force Log Rotation
root@ip-172-31-10-233:/etc/logrotate.d# sudo logrotate --force /etc/logrotate.conf
root@ip-172-31-10-233:/etc/logrotate.d# cd /var/log
root@ip-172-31-10-233:/var/log# ls -ltr /var/log
total 2296
drwx----- 2 speech-dispatcher root          4096 Apr  7 2024 speech-dispatcher
drwxr-xr-x  2 root                root          4096 Jul 25 2025 dist-upgrade
drwxr-x--  2 root                root          4096 Aug 12 2025 sssd
lrwxrwxrwx  1 root                root           39 Oct 22 2025 README ->
../usr/share/doc/systemd/README.logs
drwxr-xr-x  2 root                root          4096 Nov 24 2025 openvpn
drwxr-sr-x+ 3 root                systemd-journal 4096 Dec  3 13:16 journal
drwx----- 2 root                root          4096 Dec  3 13:16 private
drwxr-x--  2 _chrony              _chrony        4096 Dec  3 13:16 chrony
drwxr-xr-x  2 landscape            landscape    4096 Dec  3 13:17 landscape
drwxr-xr-x  3 root                root          4096 Dec  3 13:22 hp
drwxr-xr-x  2 cups-browsed         lpadmin      4096 Dec  3 13:25 cups-browsed
drwx--x--x  2 root                gdm          4096 Dec  3 13:30 gdm3
-rw-r--r--  1 root                root           0 Dec  8 07:07 fontconfig.log
-rw-r----- 1 root                adm           0 Dec  8 07:07 apport.log
-rw-r--r--  1 root                root           0 Dec  8 07:07 ubuntu-advantage-apt-hook.log
drwxr-xr-x  2 root                root          4096 Dec 20 07:00 cups
drwx----- 3 root                root          4096 Dec 20 07:05 amazon
-rw-r--r--  1 root                root          370 Dec 20 07:05 dpkg.log.2.gz
drwxr-x--  2 root                adm          4096 Jan  2 10:56 unattended-upgrades
-rw-r--r--  1 root                root           51 Jan  2 11:00 dcv-setup.log
-rw-r--r--  1 root                root           781 Jan  2 11:17 alternatives.log.1
-rw-r--r--  1 root                root          80215 Jan  2 11:19 dpkg.log.1
-rw-r----- 1 syslog              adm          64478 Jan  5 07:14 kern.log.4.gz
-rw-r--r--  1 root                root           0 Jan  5 07:14 alternatives.log
-rw-r----- 1 syslog              adm          114701 Jan  6 05:36 cloud-init.log.2.gz
-rw-r----- 1 root                adm          14981 Jan  7 06:42 dmesg.4.gz
-rw-r----- 1 root                adm          14765 Jan  7 09:56 dmesg.3.gz

```

|            |          |      |                                            |
|------------|----------|------|--------------------------------------------|
| -rw-r----  | 1 root   | adm  | 14968 Jan 10 10:00 dmesg.2.gz              |
| -rw-r----  | 1 root   | adm  | 14921 Jan 10 13:23 dmesg.1.gz              |
| -rw-r----  | 1 syslog | adm  | 100215 Jan 24 06:06 kern.log.3.gz          |
| -rw-r----  | 1 syslog | adm  | 6441 Jan 24 06:06 auth.log.4.gz            |
| -rw-r----  | 1 syslog | adm  | 433316 Jan 24 06:06 syslog.4.gz            |
| -rw-r----  | 1 root   | adm  | 48255 Jan 24 06:07 dmesg.0                 |
| -rw-r-r--  | 1 root   | root | 1435 May 30 03:41 gpu-manager.log          |
| drwxr-xr-x | 2 root   | root | 4096 May 30 03:41 sysstat                  |
| -rw-r----  | 1 syslog | adm  | 13057 May 30 03:41 kern.log.2.gz           |
| -rw-r----  | 1 syslog | adm  | 1499 May 30 03:41 auth.log.3.gz            |
| drwxr-xr-x | 2 root   | root | 4096 May 30 03:41 apt                      |
| -rw-r-r--  | 1 root   | root | 0 May 30 03:41 dpkg.log                    |
| -rw-r----  | 1 syslog | adm  | 54720 May 30 03:41 syslog.3.gz             |
| -rw-r----  | 1 root   | adm  | 50200 May 30 03:42 dmesg                   |
| -rw-r----  | 1 syslog | adm  | 89 May 30 03:42 kern.log.1                 |
| drwxrwsrwt | 2 dcv    | dcv  | 4096 May 30 03:42 dc                       |
| -rw-r----  | 1 root   | adm  | 49747 May 30 03:42 cloud-init-output.log.1 |
| -rw-r----  | 1 syslog | adm  | 1065239 May 30 03:42 cloud-init.log.1      |
| -rw-rw-r-- | 1 root   | utmp | 292292 May 30 03:44 lastlog                |
| drwxr-xr-x | 2 root   | root | 4096 May 30 03:44 sessions                 |
| -rw-r----  | 1 syslog | adm  | 1128 May 30 03:50 auth.log.2.gz            |
| -rw-rw---- | 1 root   | utmp | 0 May 30 03:50 bttmp.1                     |
| -rw-r----  | 1 syslog | adm  | 0 May 30 03:50 kern.log                    |
| -rw-r----  | 1 syslog | adm  | 24783 May 30 03:50 syslog.2.gz             |
| -rw-rw-r-- | 1 root   | utmp | 0 May 30 03:50 wttmp.1                     |
| -rw-r----  | 1 syslog | adm  | 664 May 30 03:56 auth.log.1                |
| -rw-rw---- | 1 root   | utmp | 0 May 30 03:56 bttmp                       |
| -rw-r----  | 1 syslog | adm  | 577 May 30 03:56 syslog.1                  |
| -rw-rw-r-- | 1 root   | utmp | 0 May 30 03:56 wttmp                       |
| -rw-r----  | 1 syslog | adm  | 109 May 30 03:56 auth.log                  |
| -rw-r----  | 1 syslog | adm  | 897 May 30 03:56 syslog                    |

root@ip-172-31-10-233:/var/log# #1. Confirmation of Rotation Rules (weekly, rotate 4, create) ☐ Yes, logs are being rotated correctly according to the configuration: Weekly: Rotation has occurred. The newest rotated files (.1, .2, etc.) are dated May 30, matching a recent weekly rotation cycle. Rotate 4: Perfectly followed. For key logs like auth.log, syslog, and kern.log, you can see exactly 4 archived versions: .1, .2.gz, .3.gz, .4.gz Create: Working correctly. New active log files (auth.log, syslog, kern.log) exist with very small sizes (109, 897, and 0 bytes respectively) and today's timestamp, showing they were freshly created after rotation.

root@ip-172-31-10-233:/var/log# #2. Compression Status (.gz files) Compression is active, even though #compress is commented out in the main /etc/logrotate.conf. Many rotated logs are compressed: auth.log.2.gz, syslog.4.gz, kern.log.4.gz, etc.

root@ip-172-31-10-233:/var/log#

root@ip-172-31-10-233:/var/log# #CONCLUSION:I have a solid understanding of how to manually force log rotations and verify them using terminal commands. This fundamental skill is crucial for me to maintain efficient log file management and ensure that system logs do not consume excessive disk space.

root@ip-172-31-10-233:/var/log#

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root@ip-172-31-10-233:/var/log# #LabComplete  
root@ip-172-31-10-233:/var/log#
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